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Steady Plio-Pleistocene diversification and a 2-million-year sympatry threshold in a New Zealand cicada radiation

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ABSTRACT

Estimation of diversification rates in evolutionary radiations requires a complete accounting of cryptic species diversity. The rapidly evolving songs of acoustically signaling insects make them good model organisms for such studies. This paper examines the timing of diversification of a large (30 taxon) group of New Zealand cicadas (genus *Kikihia* Dugdale). We use Bayesian relaxed-clock methods and phylogenetic trees based on nuclear and mitochondrial DNA data, and we apply alternative combinations of evolutionary rate priors and geological calibrations. The extant *Kikihia* taxa began to diversify near the Miocene/Pliocene boundary around the time of increased mountain-building, and both the mitochondrial and nuclear-gene trees confirm early splits of lineages currently represented by lowland forest-dwelling taxa. Most lineages originated in the Pleistocene, and sustained diversification occurred rapidly at over 0.5 lineages/my, a rate comparable to that of the Hawaiian silverswords. Diversification rate tests suggest an increase in the early to mid-Pliocene, followed by constant diversification from the Late Pliocene onward. No descendants of the many Pleistocene-age splits have evolved the ability to coexist in sympatry, and, where they do come into contact, hybrid zones have been documented based on acoustic and DNA evidence. In contrast, lineages separated in time by approximately 2 Myr often overlap in distribution with no evidence of hybridization. This suggests that at least 2 Myr has been required to achieve the level of divergence required for reproductive isolation.

1. Introduction

Many links have been proposed between climate shifts and changes in patterns of biological diversification (e.g., Gibbs et al., 2006; Rosenzweig, 1997). Directional and/or cyclical changes in temperature or precipitation may fragment habitats and spatially isolate populations (Gavrilets et al., 1998; Winkworth et al., 2005), facilitating their genetic divergence and subsequent adaptation to new climates and habitats. Alternatively, rapid climate change can cause extinction when habitats become fragmented or disappear entirely.

Recent efforts to identify the signature of climate shifts on diversification rates (the sum of speciation and extinction) have focused on the Pleistocene Epoch. During this period (1.8 Ma-present), global climates became sharply cooler and drier, and orbitally-driven (Milankovitch) oscillations changed from a historically dominant 20–40 ky periodicity to extraordinary 100 ky shifts between cold glacial and warm interglacial phases (Zachos et al., 2001). Advancing continental glaciers amplified these changes worldwide (Ruddiman, 2001). Many studies have sought correlations between Pleistocene-age climatic events, especially the Late Pleistocene climate shifts, and diversification patterns measured on time-calibrated molecular phylogenies and lineage-through-time (LTT) plots (Barraclough and Nee, 2001).

Despite the extraordinary climatic history of the Pleistocene, the evidence for correlated changes in diversification rates remains equivocal. Some studies have found evidence for increased diversification (e.g., Turgeon et al., 2005), while others suggest a downturn in diversification in the last million years (e.g., Good-Avila et al., 2006; Zink et al., 2004; Zink and Slowinski, 1995). Some studies investigating presumed Pleistocene-age taxa have found divergences dating to the Pliocene instead. A recent summary by Bennett (2004) stated that "The Quaternary [2.4 Ma to present] seems to have been an active period for population separation, including speciation...but the importance of this period relative to others is uncertain".

Estimation of diversification rates in evolutionary radiations, especially recent ones, is improved when efforts are made to include all cryptic taxa (Kozak et al., 2005). Barraclough and Vogler (2002) showed that more thorough taxon sampling increased estimates of recent diversification rates because unsampled taxa tend

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to be cryptic and recently diverged. Avise and Walker (1998) demonstrated previously undetected Pleistocene diversification by examining morphologically cryptic mtDNA "phylogroups" instead of formally described bird species. However, the use of mtDNA phylogroups requires the assumption that such clades ultimately become the morphologically and behaviorally diagnostic species-level taxa we recognize today (see also Barraclough and Vogler, 2002). Furthermore, the taxonomic application of the biological species concept may vary across groups, leading to differences in the degree to which recent diversification is represented. All of these factors can affect estimates of recent diversification rates.

This paper examines the pattern of diversification of a genus of New Zealand (NZ) cicadas (*Kikihia*) comprising approximately 30 taxa (e.g., Online Fig. B1). Acoustically signaling insects like cicadas use conspicuous songs for mate recognition; these songs evolve rapidly and thereby facilitate the detection of morphologically cryptic forms (e.g., Marshall and Cooley, 2000; Shaw, 2000) as well as interspecific hybridization (Online Fig. B2). Because songs play a central role in mate recognition, the cryptic taxa recognized are likely to represent an early stage in population isolation, divergence, and speciation (Alexander, 1964; West-Eberhard, 1983). Furthermore, in *Kikihia*, early stage speciation revealed by song divergence has been shown to correlate directly with mtDNA phylogeographic structure (unpublished data). The species concept we use for the purpose of this paper is a combination of the phylogenetic species concept (estimated using DNA data) and the biological species concept (assuming that song differences that correlate with DNA phylogroups are evidence for at least the early stages of reproductive isolation). Our emphasis is not on diagnosing "good species" in *Kikihia*, but on studying evolutionary diversification of population lineages, including early stages of diversification that may precede the diagnosis of species status by one or more criteria.

Speciation involves not just lineage splitting but the acquisition of reproductive isolation through ecological, behavioral, physiological, and reproductive divergence. NZ cicadas that are parapatric or sympatric offer natural experiments in reproductive isolation. Many closely related *Kikihia* taxa are parapatric and form hybrid zones (C. Fleming and J. Dugdale, unpublished letters and notes; Marshall et al. unpublished data), while others are sympatric and only rarely if at all form hybrids (Fleming, 1984; Lane, 1995), presumably representing a more mature stage in the speciation process. Still other *Kikihia* species are allopatric, and their degree of reproductive isolation is never tested in nature. By using songs to estimate species' ranges, we can accumulate a detailed picture of the spatial relationships of all *Kikihia* taxa. By mapping these geographic relationships onto a LTT (lineage-through-time) plot, we can further "calibrate" *Kikihia* diversification on a time-scale related to the speciation process.

The New Zealand archipelago is a useful setting for studies of the role of Pliocene and Pleistocene landscape and climate changes in diversification (Cooper and Millener, 1993). The region experienced an acceleration of mountain-building (Batt et al., 2000) around 5 Ma that coincided with early- to mid-Pliocene global cooling (Zachos et al., 2001). The rise of the Southern Alps gradually created new alpine habitats as well as striking moisture contrasts (McGlone, 1985) between the wet western coast of the South Island and other parts of NZ that range from mesic to semiarid. Thus, in NZ we can also compare diversification rates in a Pliocene plus early Pleistocene phase characterized by persistent cooling/drying and comparatively low-intensity climatic oscillation (Jansson and Dynesius, 2002) to diversification rates from a mid-to-late-Pleistocene phase characterized more by high-amplitude 100 ky climate cycles.

Fleming (1979) suggested that a major burst of speciation in *Kikihia* occurred during Pleistocene glacial cycles. Later phylogenetic analysis of 18 *Kikihia* species (Arensburger et al., 2004b) indicated that speciation increased around 3.8 Ma and that most lineages appeared before the Pleistocene. However, many cryptic taxa have since been identified and collected through acoustical and mtDNA analysis of the variable *Kikihia muta* grass-cicada group. With all 30 acoustically distinguishable species and subspecies now sampled, a complete analysis is feasible.

2. Materials and methods

2.1. Specimen and DNA data collection

Cicadas were identified by song phenotype (unpublished data and Fig. 1) and preserved in 95% EtOH at −20 to −80 °C (Table 1). The 30 sampled *Kikihia* species included fourteen undescribed taxa, ten of which were proposed by NZ naturalists C. Fleming and J. Dugdale (unpublished notes). Temporary species identifiers (in quotes) are used for these while taxonomic description is pending. Two taxa lacking diagnostic songs (*K.* "balaena" and *K.* "muta east") were included based on morphological data suggested by Fleming and Dugdale and/or genetic data derived from ongoing phylogeographic analyses (Marshall et al. unpublished data). Two or three specimens from different localities were sequenced for most species. Outgroup taxa were selected from the endemic NZ genera *Maoricicada* (*M. cassiope*) and *Rhodopsalta* (*R. microdora*), which form an exclusive monophyletic group with *Kikihia* (Arensburger et al., 2004a; Buckley et al., 2002).

Genomic DNA was extracted from leg or thoracic muscle tissue using the Qiagen DNEasy and Clontech Tissue kits. Selected nuclear (EF-1α) and mitochondrial (COI, COII, ATPase6, ATPase8) genes were amplified using the polymerase chain reaction and the Takara ExTaq kit. Primers used (5′–3′) were as follows: COI—C1-J-2195 and TL2-N-3014 (Simon et al. 1994); COII—TL2-J-3034 and TK-N-3786 (Simon et al. 1994); A6A8—GTAATTAACAAATACTCTTAATGG and GATGTCCAGCAATTATATTAGCTG; EF-1α segment 1—TGCTG CTGGTACTGGTGAAT and ACACCAGTTTCAACTCTGCC; EF-1α segment 2—CAGGAYGTATACAAAATTGGTGG and TTGATAGACTTGGG ATTTTC. The EF-1α segments overlap and encompass a region of the gene containing six exons and five introns (see Arensburger et al., 2004a). Annealing temperatures (°C) were as follows: COI—57; COII—53; A6A8—55, EF-1α—59. Amplified products were cleaned using the Qiagen PCR Purification Kit or Clontech Extract Kit. Cycle sequencing was conducted using the Applied Biosystems Big Dye Terminator v1.1 kit, and the product was cleaned by Sephadex (Millipore) filtration and visualized on an Applied Biosystems ABI 3100 capillary sequencer. Ambiguous base calls (e.g., due to possible heterozygosity in EF-1α) were coded as ambiguities in later analyses.

2.2. Sequence alignment and model selection

Sequences were aligned in Sequencher v3.1 (Gene Codes Corp., Ann Arbor, MI) and checked by eye. Gaps in EF-1α were rare and alignment was unambiguous. No mitochondrial length variation was observed, and the mtDNA sequences were combined for all analyses. Base composition homogeneity, across all sites and across variable sites only, was tested for each dataset using Paup*4.0b10 (Swofford, 1998). Because in preliminary phylogenetic analyses all conspecific samples formed either monophyletic or paraphyletic (two cases) clades, a single haplotype was selected as the exemplar for each taxon in the final analyses (see Table 1). This simplified the final trees for use in analyses of diversification rates.

Sequence evolution models for Bayesian and ML analyses were evaluated using Modeltest version 3.7. It is not yet clear what method is best for choosing sequence evolution models, so we

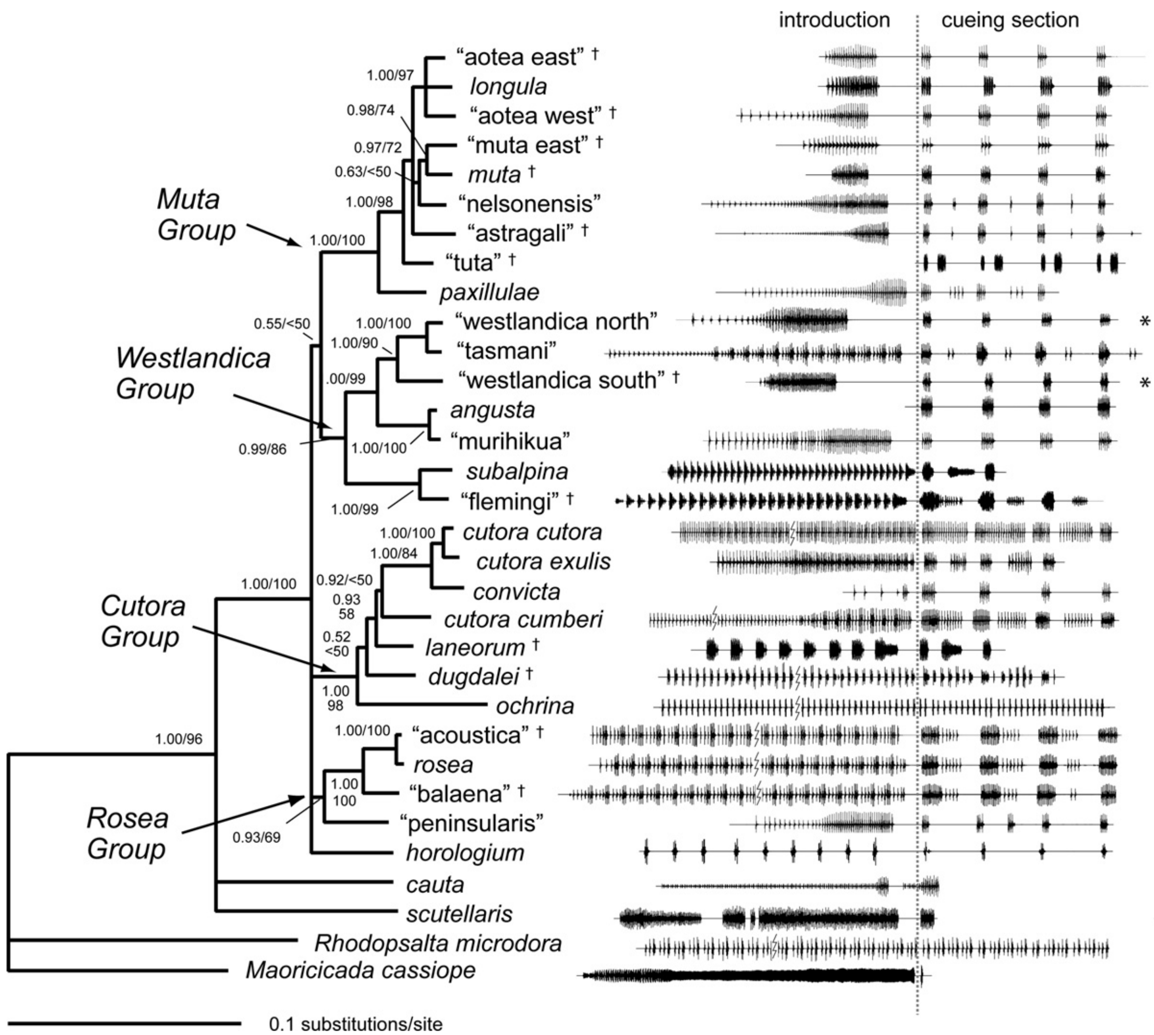

Fig. 1. Partitioned Bayesian phylogeny (post-burnin consensus phylogram) of the genus *Kikihia* based on 2152 bp of mtDNA sequence and showing taxon-specific song types (oscillograms). Major well-supported monophyletic groups mentioned in the text are noted. Bayesian posterior probabilities (first) and ML bootstrap percentages are shown for each node. Taxa added since the analysis of Arensburger et al. (2004b) are denoted with the symbol "†". Oscillograms are aligned by the division between the song introductory section and the song cueing section, which triggers female wing-flick replies (unpublished data), and by the first four song cues (some songs contain additional notes between the cues). Cueing section is truncated for most species. Some songs contain no introduction (e.g., *Kikihia angusta*), while a few (marked by large asterisks) are characterized by long cueing phrases commonly lasting 20 s or longer. Time scales are not uniform across oscillograms.

applied three of the most commonly employed model selection criteria, the likelihood-ratio test (Goldman, 1993), the Akaike Information Criterion (AIC: Akaike, 1973), and the Bayesian Information Criterion (BIC: Schwarz, 1978). The likelihood-ratio test directly compares (nested) models without consideration of model parameterization, while both the AIC and BIC explicitly attempt to penalize models containing more parameters (Burnham and Anderson, 1998). The AIC and BIC do not require models to be nested. For all datasets and partitions we selected the highest-ranked model in Modeltest that matched one of the available models in MrBayes version 3. These included the JC69 (Jukes and Cantor, 1969), K80 (Kimura, 1980), F81 (Felsenstein, 1981), HKY85 (Hasegawa et al., 1985), and GTR (Yang, 1994a) models with no among-site rate variation (ASRV), with an invariant sites parameter (Hasegawa et al., 1985), with ASRV approximated by a discrete gamma distribution (Yang, 1994b), or with both invariant sites and gamma-distributed ASRV (Gu et al., 1995). If one of the model selection criteria ranked a different model as best, the model indicated by the other two criteria was selected for our analyses.

Model-fitting was conducted with the whole mtDNA or EF-1α dataset for the ML analyses (which, in Paup* 4.0, do not allow data partitioning) and separately for the different codon positions or noncoding (EF-1α) segments for the Bayesian analyses (which allow partitioning). Partitioning protein-coding datasets by codon position often dramatically improves the log-likelihood fit of the model to the data (Brandley et al., 2005), and this effect was clearly observed in early testing of alternative partitioning strategies of our dataset (unpublished observations). Preliminary trees for model-fitting were obtained in Paup*4.0b10 using a heuristic parsimony search with 10 random addition-sequence replicates, TBR branch-swapping, and gaps coded as missing data. Because the initial Bayesian analyses indicated high dependence on the priors for 1st and 2nd position EF-1α and mtDNA partitions (presumably due to the small number of variable sites), 1st and 2nd codon positions were combined for each of the nuclear and mtDNA datasets. The few mtDNA noncoding sites (tRNA-Asp) were included with the 1st and 2nd codon positions, yielding a two-partition Bayesian model. For the EF-1α dataset an additional partition was constructed for the gaps (16 indels coded as presence/absence; Appendix A), yielding a combined model of four partitions (1st plus 2nd, 3rd, noncoding, and gaps). Gaps were coded as missing data in the ML analyses. For the final Bayesian

Table 1
Specimen locality data

Specimen	Code	Lat.	Lon.	Location
K. "acoustica"	98.MK.LOH.63	−44.2373	169.8228	0.6 km S. of Lake Ohau lodge
K. angusta	01.OL.INV.07	−44.7312	168.4560	Rees R. Vly Rd. at Invincible Mine Rd.
K. angusta [a,b]	98.MB.LSG.59	−42.1367	172.9117	Lake Sedgemere, Marlborough
K. "aotea east"	01.WN.WNU.A	−41.2493	174.9212	Wainuiomata Hill track nr. Tawa Track
K. "aotea east" [a]	02.BP.WAR.01	−38.3049	177.3956	SH2, 7.0 km S. of Wairata
K. "aotea west" [a]	01.TK.ERS.01	−39.3126	174.1464	Pembroke Rd., 8 km W. Cardiff Rd.
K. "aotea west"	02.HB.SSA.01	−39.2147	176.6883	SH5 19.4 km NW Glengary Rd., Hawkes B.
K. "aotea west" [b]	01.TO.RCG.01	−39.1919	175.5317	SH48, 0.5 km N. of Whakapapa
K. "astragali" [a]	02.NN.KNH.01	−40.6376	172.5634	Knuckle Hill Summit, NW Nelson
K. "astragali"	02.NN.KNH.03	−40.6376	172.5634	Knuckle Hill Summit, NW Nelson
K. "balaena" [a,b]	02.KA.WBH.01	−42.4950	173.1828	SH70, 3.3 km N. Lyford Lodge, Kaikoura
K. "balaena"	02.KA.WBH.03	−42.4950	173.1828	SH70, 3.3 km N. Lyford Lodge, Kaikoura
K. cauta	94.WN.HAW.72	−41.3250	174.7300	Hawkins Hill, Wellington
K. cauta [a,b]	94.WN.RIM.76	−41.1146	175.2321	SH2, Rimutaka Summit
K. convicta	98.NF.NFI.08			Norfolk Island
K. cutora cumberi [a,b]	02.TO.TPP.05	−38.6907	176.0675	Park in Taupo by Lake Taupo
K. cutora cumberi	94.WN.RIM.95	−41.1150	175.2333	Rimutaka Summit Trail
K. cutora cutora	00.AK.HAT.01	−36.5667	174.6950	Sun Vly. Rd. at Hatfield's Beach, Orewa
K. cutora cutora [a,b]	93.AK.BUL.70	−36.5000	174.6883	Bullock Track, Mahurangi West
K. cutora exulis	98.KE.RAO.46	−29.2483	178.0767	Raoul, Kermadec Islands
K. dugdalei	00.WN.DAY.02	−41.2783	174.9167	Days Bay, Wellington
K. dugdalei [a,b]	02.BP.CRE.02	−38.1542	176.2640	SH38 at Rotorua Crematorium and Cem.
K. "flemingi"	02.MB.PAT.04	−41.5895	173.2988	W. of Lk. Chalice Lookout., Mt. Patriarch
K. "flemingi" [a]	98.OL.TWE.51	−45.0685	168.5458	Twelve Mile Delta, Lake Wakitipu
K. horologium	93.MK.SEB.80	−42.2500	172.8833	Mt. Sebastopol, Mt. Cook NP
K. horologium [a,b]	97.MB.ENC.77	−41.5800	173.2567	Above Enchanted Lookout, Lake Chalice
K. laneorum [a,b]	02.TO.OPE.01	−38.7687	176.2178	SH5 at Opepe Historic Preserve
K. laneorum	02.TO.WWS.01	−38.8949	175.4960	16 km W. SH4/SH41 Jct., W. Waituhi Sdl.
K. longula	94.CH.CHA.67	−41.1067	172.6917	Chatham Islands
K. "murihikua" [b]	94.FD.RDS.01			SH94, Fiordland
K. "murihikua" [a]	01.CO.CRA.01	−44.9037	168.9849	SH89, 5.2 km SW of Cardrona
K. muta	01.WI.FER.03	−40.2299	175.5716	SH54, 0.7 km E. of Feilding Town Centre
K. "muta east" [a]	02.GB.NUH.01	−39.0442	177.7377	SH2 at Nuhaka
K. "muta east"	02.HB.GGR.02	−39.3501	176.7369	Glengarry Rd., 1.4 km from SH5 Jct.
K. "nelsonensis"	01.NN.WRR.01	−39.8344	175.1619	SH4, 3.9 km N. Wanganui River Rd.
K. "nelsonensis" [a,b]	01.SD.CUL.02	−41.2738	173.7879	Up track S. of Cullens Pt. across SH6
K. ochrina [a,b]	00.WN.DAY.01	−41.2783	174.9167	Days Bay, Wellington
K. ochrina	94.WN.NEV.03	−41.3020	174.8292	164 Nevay Rd., Miramar, Wellington
K. paxillulae	97.KA.PPR.81			Puhi-Puhi Res. at Hapuku R., Kaikoura
K. "peninsularis" [a]	98.MC.LEV.50			Rd. to Port Levy, Banks Peninsula
K. "peninsularis" [b]	01.MC.BPT.11	−43.7800	172.7882	Rd. to Te Oka, 0.8 km from SH75
K. rosea	98.DN.BBY.53			Broad Bay, Dunedin
K. scutellaris [a,b]	94.WN.QEP.93			Queen Elizabeth Park, Paekakariki
K. scutellaris	97.TO.OPE.60	−38.7687	176.2178	SH5 at Opepe Historic Preserve
K. "westlandica north" [a,b,c]	02.BR.IRO.10	−41.7867	172.0310	SH6, Iron Bridge, Buller R.
K. "westlandica north" [c]	02.BR.IRO.14	−41.7867	172.0310	SH6, Iron Bridge, Buller R.
K. "westlandica south" [a,b]	02.BR.RUN.03	−42.4127	171.2491	SH6, 0.5 km S. of Runanga at memorial
K. "westlandica south"	02.NC.APV.01	−42.9466	171.5637	Railway Stn., Arthurs Pass Village
K. subalpina	01.TO.TSR.16	−39.2963	175.7354	Tukino Skifield Rd. ca. 0.3 km W. of SH1
K. subalpina [a,b]	01.WN.RIM.01	−41.1150	175.2333	Rimutaka Summit Trail
K. "tasmani" [a]	96.NN.SYL.10			NW Nelson Trail, Lake Sylvester
K. "tasmani" [b]	02.NN.COR.21	−41.1071	172.6921	Cobb Ridge, overlooking reservoir
K. "tuta"	01.MB.TWI.03	−41.3378	173.7606	Twidles Rd., 0.2 km W. of SH6
K. "tuta"	01.NN.WCR.02	−40.5799	172.6274	12 km N. of rd. to Knuckle Hill Trail
K. "tuta" [a]	02.NN.DEB.01	−41.1798	173.4294	1.7 km N. Cable Bay Rd on Maori Pa Rd.
K. "tuta" [b]	01.NN.COL.03	−40.6810	172.6707	Jct. of SH60/Poplar Ln. nr. Collingwood
Maoricicada cassiope	*97.TO.BRR.01*	*−39.2336*	*175.5447*	*SH48, 5.5 km S. of Chateau, Mt. Ruapehu*
Rhodopsalta microdora	*02.MC.LCR.02*	*−43.6427*	*172.4781*	*Landcare Research grounds, Lincoln*

GPS data were not taken at all sites.
[a] Used as exemplar in final mtDNA tree and in combined-data analysis.
[b] Used in EF-1a tree.
[c] Referred to by the code NWCM in Arensburger et al. (2004b).

analyses a GTR + I + Γ model was applied to each of the two mtDNA partitions, with the default setting of four rate categories in each case. JC was used for the nuclear 1st and 2nd positions, HKY was used for the nuclear 3rd positions and noncoding positions, and the Mk model (Lewis, 2001) was used for the Ef-1α gap partition. For the final ML analyses (which do not allow data partitioning) a GTR + I + Γ (four category) model was used for the Ef-1α dataset and a GTR + I + Γ (four category) model was used for the mitochondrial dataset.

2.3. Phylogenetic analysis

The mitochondrial and nuclear-gene datasets were analyzed separately using both Bayesian and maximum-likelihood methods, after which a partitioned Bayesian combined-data analysis was conducted with six data partitions as described above. No combined-data ML analysis was conducted because of the substantial differences observed between the parameter estimates of the separate nuclear and mitochondrial datasets.

Bayesian analyses using MrBayes v3.1 (Ronquist and Huelsenbeck, 2003) were run twice, with *nruns* set to 2 each time, for ten million generations with four chains (three heated). Exponential priors were selected for the α shape parameter prior (exp[.1]) and branch lengths (mtDNA dataset: exp[75]; EF-1α dataset exp[500]); the latter values were selected to approximate the ML estimates of total tree length. Among-partition rate variation was accommodated by setting *ratepr=variable* (see Marshall et al., 2006). All model parameters except branch lengths and topology were unlinked across partitions. Default settings were used otherwise. A burn-in of approximately 33% was sufficient for all analyses. Stationarity was confirmed by (1) checking for stability of post-burnin mean log-likelihood and parameter estimates in the program Tracer v1.4 (Rambaut and Drummond, 2003), (2) checking that the uncorrected potential scale reduction factors (PSRF: Gelman and Rubin, 1992) remained near 1.0 with the MrBayes *nruns* setting on 2, and (3) confirming that the average standard deviation of split frequencies remained below 0.01 for all post-burnin samples. Tracer v1.4 was also used to confirm the adequacy of the posterior sample size by examination of sample autocorrelation. MrBayes posterior probabilities (PP) were used to assess nodal support.

ML analyses were run in Paup* 4.0b10 (Swofford, 1998) using default settings except as noted below. Model parameters were estimated on a preliminary NJ tree and then used in a heuristic ML search with five random addition-sequence replicates. Parameters were re-estimated on the resulting tree and the process was repeated until the parameters stabilized (Sullivan et al., 2005). The base composition parameters were then fixed in a more thorough ML heuristic search with five random addition-sequence replicates. Nodal support (bootstrap percentage, or BP) was estimated by a nonparametric bootstrap of 100 pseudoreplicates, each involving a single heuristic search using the final ML parameter estimates.

2.4. Molecular clock test

All relaxed-clock analyses were conducted on the mtDNA tree. This choice allowed us to take advantage of the superior resolution of that tree as well as published estimates of insect mtDNA substitution rates (see below). To determine if a strict molecular clock model (Felsenstein, 1985; Zuckerkandl and Pauling, 1965) was appropriate for dating the mtDNA phylogeny, we conducted a likelihood-ratio test comparing the log-likelihood of the final Bayesian mtDNA topology (minus the outgroup taxon) with and without the clock constraint and T-2 degrees of freedom, where T = number of taxa, using PAUP* (Swofford, 1998). A second test was then conducted with the long *K. ochrina* branch removed.

2.5. Divergence time estimation

Bayesian divergence time estimation under a relaxed molecular clock was conducted using the MULTIDISTRIBUTE package of Thorne and Kishino (2002) and the Bayesian mtDNA topology, rooted on *Rhodopsalta microdora*. In preliminary exploration of the data, we attempted to use the r8s package (version 1.71) of Sanderson (2002), but the results from this program were highly variable, and the essential "cross-validation" procedure offered by the package would not complete normally, probably in part due to the lack of a strong date calibration for the root of the tree (r8s manual p. 27). Branch lengths for each mtDNA partition were optimized separately using baseml (in PAML v. 3.14; Yang, 1997) and estbranches according to the MULTIDISTRIBUTE software instructions. MULTIDIVTIME analyses were run with one million generations of burnin and sampled 10,000 times over two million generations. Repeated runs were compared to ensure stationarity.

Table 2
Prior distribution means for the relaxed-clock dating trials

Analyses	Rate type	Rate (rtrate)	Ingroup Age (rttm)	ν
C, F	Fast (Gillooly)	0.0165	5.6	0.27
B, E	Medium (Brower)	0.0115	8.0	0.18
A, D	Slow	0.008	11.5	0.13

Each analysis was conducted separately without sequence data to estimate the prior distributions of the date estimates.

MULTIDIVTIME requires the selection of means and standard deviations for priors on the substitution rate (rtrate), ingroup root age (rttm), and Brownian motion constant (ν). The choice of root substitution rate prior can influence the outcome (e.g., Buckley and Simon, 2007), so we conducted three analyses based on different rtrate priors chosen to bracket the likely true root substitution rate (see below and Table 2). For each analysis, we estimated the rttm prior by dividing the average pairwise distance through the ingroup node on the mtDNA ML tree by the assumed root substitution rate. We then set ν such that rttm * ν = 1.5, as suggested in the MULTIDIVTIME manual.

The three root substitution rate priors were selected as follows. A "medium" substitution rate test was conducted by setting rtrate to 0.0115 substitutions/site/Myr, the arthropod mtDNA estimate of Brower et al. (1994). Gillooly et al. (2005) suggest that mtDNA substitution rates are being underestimated for many organisms, and they propose a "metabolic clock" model based on body size and temperature. We therefore tested a "fast" substitution rate (0.0165 s/s/Myr) obtained by applying typical estimates of *Kikihia* body size (0.25 g) and average New Zealand lowland temperature (≈12.5 °C; http://www.niwascience.co.nz/edu/resources/climate/overview/) into Equation 2 of Gillooly et al. (2005). Lastly, we tested a "slow" substitution rate (0.008 s/s/Myr) that reduced the Brower clock correspondingly—this estimate is closer to rates found in two recent surveys of vertebrates (Pereira and Baker, 2006) and invertebrates (Quek et al., 2004). The resulting priors calculated for rttm and ν are shown in Table 2. Standard deviations for all priors were set to equal the mean, allowing for considerable uncertainty. The upper boundary on root age (bigtime) was set to 30.0 Ma (corresponding to an Oligocene submergence of much of NZ; Cooper and Cooper, 1995), and the values for the minab (1.0), newk (0.1), and othk (0.5) parameters (which influence the Markov chain proposals) were left unchanged as recommended. Minab governs the prior distribution of divergence times, while newk and othk influence the Metropolis-Hastings proposals. "Commonbrown" was set to 0 for all analyses to allow rates to be estimated separately for each partition.

2.5.1. Geological calibrations

Because fossil data are unavailable for New Zealand cicadas, geological events were used to calibrate two nodes on the mtDNA tree. First, the maximum age of the node supporting *K. convicta* was set to 3 Ma, the approximate time at which volcanic Norfolk Island became habitable (McLoughlin, 2001). Norfolk Island is the only habitable land mass known for that region historically. Second, the maximum age of the *K.* "flemingi"/*K. subalpina* split was set to 1.2 Ma, the approximate time at which North Island subalpine habitats (which support *K. subalpina*) appeared during increased uplift and volcanism (Raven, 1973; Rogers, 1989). All analyses included these two geological constraints.

Because the nearest populations of *K.* "flemingi" are found just across Cook Strait in northern South Island, we considered the possibility that the *K. subalpina* ancestor would have become established on the North Island comparatively soon after appropriate habitat first appeared, rather than more recently. In order to examine the effect of such an assumption while still providing a range of possible dates for the node, we conducted an independent set of three analyses (Analyses D–F) with the minimum age of the *K. subalpina* constrained to >800 ky in addition to the other calibrations. *A priori*, we considered the three assumptions combined (Analysis E; see Section 3) to represent the "best guess" set of prior calibrations.

2.5.2. Diversification rates and lineage-through-time (LTT) plots

To facilitate comparison of the *Kikihia* radiation with other studies, two estimates of the per-lineage diversification rate (species per lineage per Myr) were calculated, the Kendall-Moran estimator *S* (Baldwin and Sanderson, 1998) with Moran variance as advocated by Nee (2001) and a simpler statistic, *D* (Good-Avila et al., 2006) originally developed by Vrba (1987). These were calculated separately for each of the six analyses, both for the entire genus and for the "main radiation" (see Section 3). *S* is a more accurate estimator (Baldwin and Sanderson, 1998), but many studies have used only *D* (which requires no phylogenetic information). *S* is defined as $(N-2)/B$, where *N* is the number of extant lineages and *B* is the summed branch lengths (time units on a chronogram) connecting those lineages to their common ancestor. *D* is defined as $[\ln(N_1)-\ln(N_0)]/T$, where N_0 and N_1 are the number of lineages preceding and following a time interval of *T* millions of years.

To visualize the temporal pattern of diversification, a lineage-through-time (LTT) plot (log of the number of lineages plotted against time) was constructed for each of the six dating analyses (Barraclough and Nee, 2001). Using the resulting information on the overall timing of speciation events, we used the γ statistic (Eq. (1)) of Pybus and Harvey (2000) as adapted by Barraclough and Vogler (2002) to test for constancy of diversification rate under a pure-birth (Yule) speciation model. In this equation, *m* is the number of lineages at the start of the radiation, *n* is the number of lineages at the end, and $g_m \ldots g_n$ are the time intervals between successive speciation events (over the entire radiation at once; see Fig. 1 in Pybus and Harvey (2000). A constant-rate model is rejected at $p<0.05$ if $-1.96<\gamma<1.96$ (two-tailed test; see Pybus and Harvey, 2000 for further explanation). This statistic was calculated separately from the beginning of the *Kikihia* diversification and from the beginning of the "main radiation" (see Section 3). For the purposes of this test, polytomies were resolved as sets of individual bifurcations with very short internodes.

$$\gamma=\frac{\left(\frac{1}{n-m}\sum_{i=m}^{n-1}\left(\sum_{k=m}^{i}kgk\right)\right)-\left(\frac{T}{2}\right)}{T\sqrt{\frac{1}{12(n-m)}}},\ T=\sum_{j=m}^{n}jgj \qquad (1)$$

In addition, to quantify potential contrasts in overall diversification rate associated with climate changes, *D* statistics were compared for equal-interval periods preceding and following three events: (A) the 1.8 Ma start of the Pleistocene Epoch, (B) the 1.2 Ma onset of 100 ky climate cycling as identified by Carter (2005), and (C) the 0.9 Ma onset of 100 ky cycling as identified by the global temperature proxy of Zachos et al. (2001). Because changes in *D* were examined rather than the overall magnitude, these values were calculated only for the analysis representing the "best guess" priors, the medium-clock analysis including the minimum *K. subalpina* constraint (Analysis E).

2.6. Geographic analysis

Distributional data were used to investigate the effect of time since divergence on the geographic relationships of lineages. Geographic distributions for all taxa were determined from our database of 5500 specimen and song records from over 1100 GPS-indexed locations (web-searchable database available at http://hydrodictyon.eeb.uconn.edu/projects/cicada/databases/). Two species were coded as sympatric if specimens of both were collected or heard at one or more locations, except when such "overlap" was limited to the point of geographic contact between otherwise locally non-overlapping ranges (parapatry). Species with non-overlapping ranges separated by geographic gaps were coded as allopatric. Phylogenetically related clusters of mutually parapatric and/or allopatric species were mapped on the *Kikihia* chronogram from the analysis using the "best guess" set of priors (Analysis E).

3. Results

Sequences have been deposited in GenBank (Accession Nos. COI: AF247627, EF051355–EF051385, EU717569–EU717587; COII: EF051386–EF051416, EU717588–EU717607; A6A8: EF051417–EF051446, EU717550–EU717568; EF-1α: EU717608–EU717634). For five taxa, EF-1a would not sequence cleanly: *Kikihia rosea*, *K.* "aotea east", *K.* "astragali", *K.* "flemingi", and *K.* "muta east". These taxa were therefore omitted from the combined-data analysis. No significant differences in base composition were detected for sequences within any dataset.

3.1. Phylogenetic reconstruction—mtDNA

The Bayesian (Fig. 1) and ML (not shown) mtDNA phylogenies generally confirmed the *Kikihia* tree structure found by Arensburger et al. (2004a); their Fig. 2) and resolved additional relationships. The ML tree was extremely similar to the Bayesian tree in topology and branch length reconstruction, and there were no well-supported (BP > 75%) topological differences from the Bayesian tree. The ML tree resolved all of the polytomies from the Bayesian mtDNA tree with extremely short branches, none of which received significant bootstrap support. All *Kikihia* species formed a well-supported monophyletic clade (PP = 1.00, BP = 96%), and the earliest splits from other *Kikihia* were lineages leading to the distinctive "shade singer" species *K. scutellaris* and *K. cauta*, long distinguished by their preference for deep bush habitats (Fleming, 1975; Myers, 1929). Of the remaining 28 taxa, 27 grouped into four well-supported clades arising from a large polytomy here termed the "main radiation". These are the Muta Group (mostly NI and northern SI grass species), the Westlandica Group (mostly SI grass and grass/shrub species), the Cutora Group (mainly NI bush and shrub species), and the Rosea Group (SI scrub forms). *K. horologium*, a SI subalpine species, branched off independently from the large polytomy, although the ML analysis placed *K. horologium* at the base of the Rosea Group with weak support. Most of the newly added taxa were derived from recent lineage-splitting events.

3.2. Phylogenetic reconstruction—EF-1α

The Bayesian (Fig. 2) and ML (not shown) EF-1α trees confirmed key structural features of the mtDNA tree and contradicted other relationships, although sequence divergence levels were low and more recent nodes were poorly resolved. The monophyly of *Kikihia* and the early divergence of lineages leading to *Kikihia scutellaris* and *K. cauta* were strongly supported (PP = 1.00, BP = 100%). Although few other relationships were well-resolved there was partial support for the major relationships observed in the mtDNA tree—one monophyletic clade (PP = 1.00, BP = 83%) on the EF-1α tree contains all of the species from the Muta and Cutora groups of the mtDNA tree (largely North Island), while a less well-supported monophyletic group (PP = 0.96, BP = 51%) on the EF-1α tree contains all of the species from the mtDNA Westlandica and Rosea Groups, together with *K. horologium* (mainly South Island). These NI and SI groups are not found on the mtDNA tree but are not contradicted either because the major radiation is impossible to resolve on the mtDNA tree.

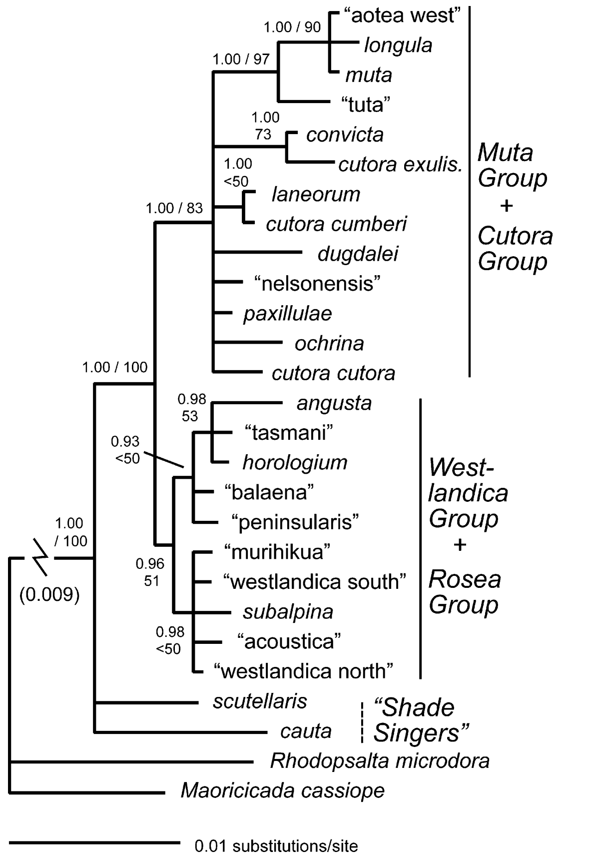

Fig. 2. Partitioned Bayesian phylogeny (post-burnin consensus phylogram) of the genus *Kikihia* based on 1545 bp of nuclear EF-1α sequence. Major well-supported monophyletic groups observed in the mtDNA tree are marked with vertical solid bars. Bayesian posterior probabilities (first) and ML bootstrap percentages are shown for each node.

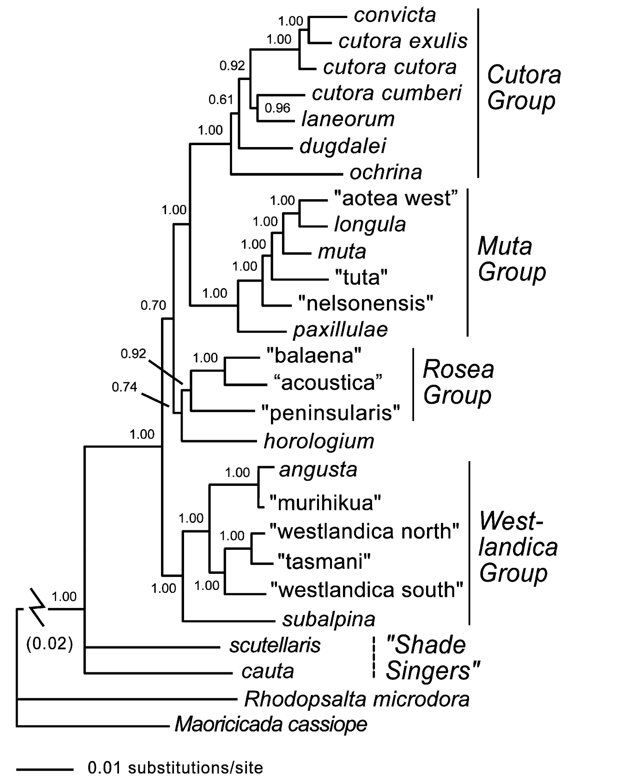

Fig. 3. Partitioned Bayesian phylogeny (post-burnin consensus phylogram) of the genus *Kikihia* based on combined analysis of 2152 bp of mtDNA sequence and 15-45 bp of nuclear EF-1α sequence. Major well-supported monophyletic groups mentioned in the text are marked with vertical solid bars. Bayesian posterior probabilities are shown for each node.

At the level of closer relationships, significant conflict was observed between the EF-1α and mtDNA trees, particularly over the positions of *K.* "acoustica", *K. angusta*, *K.* "tasmani", and *K.* "nelsonensis". All of these contradictory relationships involved comparatively short branches and extremely low ML bootstrap support values.

3.3. Phylogenetic reconstruction—combined-data

The Bayesian combined-data tree (Fig. 3) confirmed the four well-supported major clades observed in the mitochondrial tree, as well as the early splits of the lineages leading to *K. cauta* and *K. scutellaris*. Presumably indicating the influence of the nuclear-gene data, strong support was found for the Cutora + Muta Group clade as well (two clades that are mainly North Island limited).

3.4. Molecular clock test

The molecular clock hypothesis was rejected for both for the entire mtDNA tree (minus *R. microdora*) ($\chi^2 = 64$, df = 30, $p < 0.001$) and for the mtDNA tree with the long *K. ochrina* branch removed ($\chi^2 = 47$, df = 29, $p < 0.015$), therefore relaxed-clock estimates of divergence times were calculated.

3.5. Divergence time estimation

Divergence time estimates were influenced by the prior on the root substitution rate as well as by the particular combination of geological calibrations used. Mean divergence times (Table 3) for the root node (the *Kikihia*/*Maoricicada* split) ranged from 12.4 Ma with the slow rate prior and the minimum-age *K. subalpina* constraint (Analysis D) to 6.75 Ma with the fast rate prior and no minimum age for *K. subalpina* (Analysis C). The *Kikihia* ingroup node estimate ranged from 7.28–3.98 Ma across the six analyses, and the main *Kikihia* radiation ranged from 3.99–2.20 Ma. The number of terminal *Kikihia* lineages (out of 30) that dated to 1.8 Ma or younger ranged from 18 (60%) to 25 (83%). The 95% confidence intervals for all estimates were broad but comparable to those observed in a related analysis of the alpine NZ cicada genus *Maoricicada* (Buckley and Simon, 2007). Standard deviations (not shown) of the posterior date estimates were regularly about 40–45% of the mean.

The relaxed-clock dating analyses all reached stationarity during burnin as indicated by similar results of duplicated runs (data not shown). Prior means of the divergence times, as well as their confidence intervals, departed substantially from the posterior estimates (Table 3), indicating a reassuring effect of the data on the relative ages of the nodes.

Divergence time estimates for the two calibrated nodes fell well below the maximum bounds imposed. In the slow-clock Bayesian analysis with the minimum-age *K. subalpina* constraint, which generated the oldest date estimates, the *Kikihia convicta* split was dated at 0.84 Ma and the *K. subalpina*/"flemingi" split was dated at 0.97 Ma. The upper edge of the 95% confidence interval for the *Kikihia convicta* estimate did not approach the constraint imposed on the node, while the upper edge of the 95% confidence interval for the *K. subalpina*/K. "flemingi" split was close to the upper bound.

3.6. Lineage-through-time (LTT) plots

Kendall-Moran per-lineage (per Myr) diversification rates (S) for the genus, calculated for each of the six analyses, ranged from

Table 3
Mean divergence time estimates (Ma) of key nodes on the *Kikihia* mtDNA tree from the six relaxed-clock dating analyses

Node	Without minimum *K. subalpina* constraint			With minimum *K. subalpina* constraint		
	(A) Slow rate	(B) Medium rate	(C) Fast rate	(D) Slow rate	(E) Medium rate	(F) Fast rate
Root	10.9 (5.19–19.6)	8.81 (3.84–16.8)	6.75 (2.73–13.5)	12.4 (6.91–20.7)	11.1 (6.32–18.6)	9.94 (5.75–16.3)
Prior	*4.08 (0.09–14.98)*	*3.64 (0.14–12.94)*	*3.18 (0.11–11.13)*	*5.72 (1.53–17.40)*	*2.64 (0.68–7.60)*	*2.39 (0.67–6.55)*
Ingroup	6.39 (3.10–11.2)	5.19 (2.30–9.81)	3.98 (1.65–7.88)	7.28 (4.29–11.69)	6.56 (3.94–10.6)	5.94 (3.57–9.41)
Prior	*3.66 (0.08–13.72)*	*3.28 (0.13–11.60)*	*2.86 (0.10–10.01)*	*5.00 (1.37–15.21)*	*2.08 (0.45–6.04)*	*1.89 (0.43–5.30)*
Main radiation	3.47 (1.67–6.04)	2.85 (1.27–5.35)	2.20 (0.89–4.37)	3.99 (2.43–6.32)	3.65 (2.26–5.79)	3.36 (2.12–5.29)
Prior	*2.83 (0.06–10.34)*	*2.55 (0.10–9.06)*	*2.22 (0.07–7.82)*	*4.26 (1.23–12.97)*	*1.52 (0.27–4.33)*	*1.39 (0.24–3.91)*
K. muta radiation	1.75 (0.81–3.17)	1.44 (0.60–2.78)	1.11 (0.44–2.28)	2.01 (1.15–3.34)	1.84 (1.05–3.08)	1.69 (0.98–2.81)
Prior	*2.04 (0.04–8.12)*	*1.79 (0.05–6.95)*	*1.57 (0.04–6.04)*	*3.00 (0.79–9.31)*	*2.65 (0.77–7.80)*	*2.41 (0.73–6.74)*
K. convicta	0.72 (0.30–1.46)	0.60 (0.22–1.25)	0.46 (0.16–1.02)	0.84 (0.39–1.55)	0.76 (0.36–1.42)	0.70 (0.33–1.28)
Prior	*0.78 (0.01–2.64)*	*0.74 (0.02–2.56)*	*0.66 (0.01–2.45)*	*1.02 (0.12–2.70)*	*0.95 (0.10–2.65)*	*0.88 (0.09–2.59)*
K. subalpina	0.81 (0.38–1.18)	0.69 (0.28–1.15)	0.56 (0.20–1.09)	0.97 (0.76–1.18)	0.95 (0.76–1.18)	0.92 (0.76–1.18)
Prior	*0.43 (0.01–1.14)*	*0.42 (0.01–1.13)*	*0.40 (0.01–1.12)*	*0.96 (0.76–1.19)*	*0.96 (0.76–1.19)*	*0.96 (0.76–1.19)*

Mean age estimates for the main *Kikihia* radiation predate the onset of the Pleistocene under all combinations of rate assumptions and geological calibrations tested, while the earliest divergences of the genus approximate the Miocene/Pliocene boundary.
Note: 95% confidence intervals are given in parentheses. Values from the prior distribution are given in italics.

Table 4
Diversification rate statistics for the six relaxed-clock dating analyses

	Without minimum *K. subalpina* constraint			With minimum *K. subalpina* constraint		
Node	(A) Slow rate	(B) Medium rate	(C) Fast rate	(D) Slow rate	(E) Medium rate	(F) Fast rate
D—Genus	0.53	0.66	0.86	0.47	0.52	0.57
S—Genus	0.45 (0.01)	0.55 (0.01)	0.72 (0.02)	0.40 (0.01)	0.43 (0.01)	0.47 (0.01)
D—Main radiation	0.64	0.78	1.0	0.56	0.61	0.67
S—Main radiation	0.57 (0.01)	0.69 (0.01)	0.89 (0.03)	0.49 (0.01)	0.54 (0.01)	0.58 (0.01)

Note: Each statistic is calculated separately for the entire *Kikihia* genus and for the main radiation alone. Both assume a pure-birth Yule model of diversification. The Moran estimate of variance is given (in parentheses) for each estimate of *S*. See text for formulas.

0.40–0.72, while the values for the main radiation were somewhat faster, 0.49–0.89 (Table 4). Good-Avila *D* estimates were slightly larger in each case. The LTT plots (Fig. 4) suggested a minor increase in diversification rate early, around 3.9–2.2 Ma, at the start of the main radiation, followed by apparently steady accumulation of lineages until the present. However, γ tests did not reject the constant-rate pure-birth model for the genus as a whole ($\gamma = -0.67$) or for the "main radiation" alone ($\gamma = -1.92$), although the latter value indicates a marginally nonsignificant slowing trend. "Composite" (across lineages) *D* statistics showed no strong contrasts associated with key climatic events, with the possible exception of the 0.9 Ma contrast (0.50 versus 0.34).

3.7. Spatial relationships and divergence times

The spatial relationships of *Kikihia* taxa are strongly predicted by the time since their most recent common ancestor. Assuming the date estimates from Analysis E (Table 3), nearly all species pairs with divergence times dating to the Pleistocene exhibit parapatric or allopatric relationships (Fig. 5), while most with older divergence times are sympatric or appear potentially so (as when, for example, two species are found on different islands but the sister-taxon of one species overlaps the other). The pattern holds in five clades that have independently undergone lineage splitting events during the past 2 Ma. The inferred "sympatry threshold" is apparently deeper in one of these clades, the Rosea clade (ca. 3 Ma).

4. Discussion

4.1. Phylogenetic relationships of Kikihia and gene-tree conflict

Major features of the *Kikihia* phylogeny are well-supported by both the mtDNA and EF-1α datasets. These include the monophyly of the genus, the early splits of species-poor lineages today represented only by lowland forest-dwelling species, and a large polytomy containing at least two major groups, one mainly North Island taxa and the other mainly South Island taxa. However, significant posterior probability support was found for some conflicting relationships. Conflict between gene trees arises from many processes, including poor phylogenetic analysis (e.g., model misspecification), incomplete lineage sorting, and gene introgression following interspecific hybridization (Holder et al., 2001; Nichols, 2001). In this situation, there is evidence to suggest that interspecific hybridization may be causing one gene tree to mislead us about species-level lineages. Two closely related species pairs observed in the mtDNA tree (*K.* "tasmani" + *K.* "westlandica north", *K.* "murihikua" and *K. angusta*) have unusually divergent songs (Fig. 1) for the amount of mtDNA divergence between them. This may indicate that the mtDNA haplotype of one species has introgressed and become fixed in populations of the other species. Both species pairs have geographically proximate or overlapping ranges (Fig. 5), and intermediate song phenotypes of the latter pair have been observed in areas of overlap (unpublished data). The separation of K. "acoustica" and K. "balaena" in the EF-1α tree is also surprising in light of their extremely similar song phenotypes, but low nodal support values (especially ML bootstraps) suggest caution in interpreting these relationships. Clearly, both hybridization and lineage-sorting remain at least plausible explanations for each case of conflict, and a more careful analysis of this question will have to await additional nuclear-gene trees (e.g., Buckley et al., 2006).

4.2. Pleistocene climates and cicada diversification

As expected, the addition of the fourteen remaining species, many of them morphologically cryptic, more than doubled the fraction of Pleistocene-age *Kikihia* diversity from 28% (Arensburger et al., 2004a) to around 67%. With most of the newly added taxa

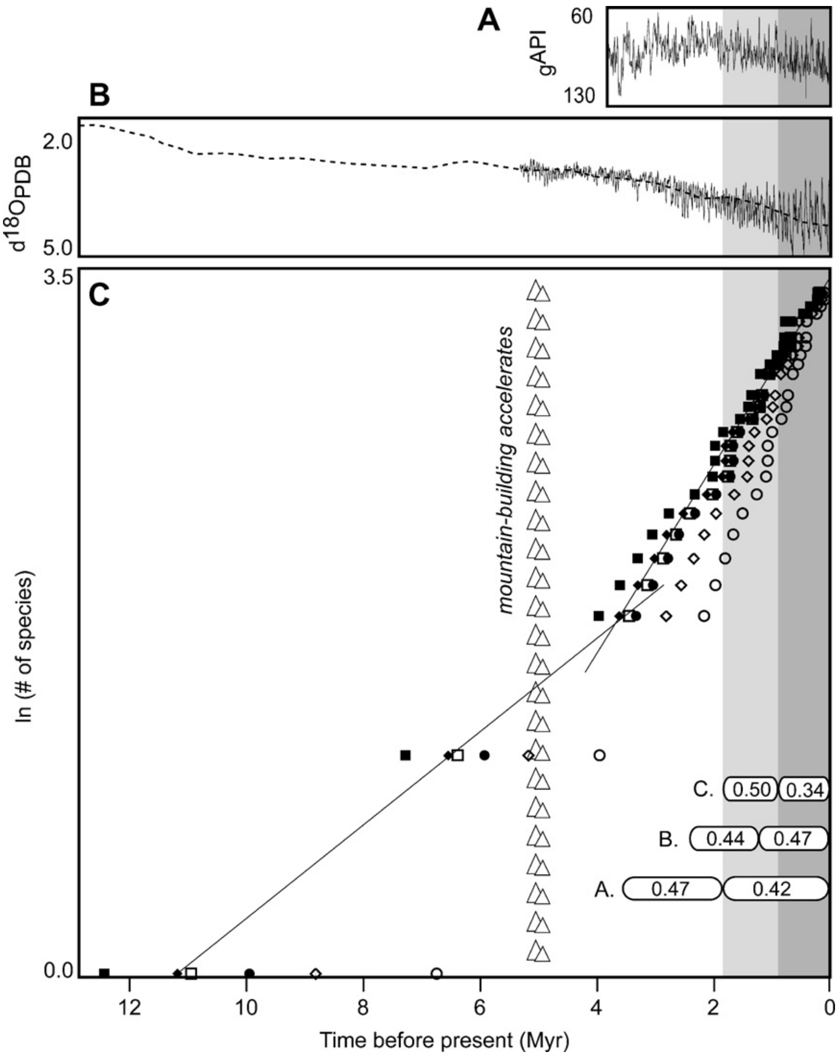

Fig. 4. Steady Late Pliocene-Pleistocene diversification pattern in the cicada genus *Kikihia* illustrated by lineage-through time plots (c) derived from relaxed-clock dating analyses of the mtDNA dataset. The plot is time-calibrated to two temperature proxies, a global temperature proxy (b—oxygen isotope) from Zachos et al. (2001) and a New Zealand temperature proxy (a—gamma-ray) from Carter (2005)—larger values of each are correlated with colder temperatures. Also shown is the estimated timing of the onset of major mountain-building on the South Island. The six different symbols used in (C) correspond to the results of the six relaxed-clock mtDNA dating analyses found in Table 3; these show the range of mean estimates observed with different root rate assumptions and geological calibrations. Divergence times from slow, medium (Brower), and fast (Gillooly) rate analyses are plotted using squares, diamonds, and circles, respectively. Filled symbols show data from the analyses employing the minimum *K. subalpina* calibration (see Section 2); open symbols represent the data from the analyses without the minimum *K. subalpina* calibration. Early (light shading) and late (dark shading) Pleistocene periods are indicated. For visual purposes, regression lines are plotted for the results of Analysis E (see Section 2) preceding and following an apparent increase in diversification around 4 Ma. Values in the lower right of the diagram are estimates of the per-lineage speciation rate per my (D: Good-Avila et al., 2006) over equal-interval periods preceding and following (A) the onset of the Pleistocene (1.8 Ma), (B) the onset of 100 ky climate cycling in New Zealand as indicated by Carter (2005) (1.2 Ma), and (C) the onset of global 100 ky Late Pleistocene climate shifts (0.9 Ma) as indicated by Zachos et al. (2001).

connecting to the tips of the tree, the pace of *Kikihia* diversification now appears greater than in prior studies. Diversification rate estimates ($S \approx 0.54$ for the main radiation in the more conservative analyses) for the *Kikihia* clade are similar to those obtained for the rapid radiations of the Hawaiian silverswords ($S = 0.56$) (Baldwin and Sanderson, 1998) and North American tiger beetles ($S > 0.22$; Barraclough and Vogler, 2002). Less conservative priors suggest estimates ($S = 0.57–0.89$) closer to those observed by Kozak et al. (2005) for the peak period of *Plethodon* salamander diversification. It is possible that the illumination of cryptic diversity by cicada song leads to higher diversification rate estimates compared to groups in which only morphology is available for delimiting taxa. The highest short-term diversification rate measured for an arthropod ($S = 4.17$ species per species per Myr) comes from a study of Hawaiian *Laupala* crickets, many of which have been described using song characters (Mendelson and Shaw, 2005).

While most *Kikihia* diversification has occurred recently, it does not appear that Pleistocene-age processes have been more important to NZ cicada evolution than species-generating mechanisms of the middle and late Pliocene. Date estimates placing the beginning of the main *Kikihia* radiation close to the onset of the Pleistocene were obtained only when a mean root substitution rate of 0.0165 s/s/Myr was assumed (Table 3), a rate half again as fast as the "classical" mtDNA clock of Brower (1994). Furthermore, for the past 2.5–4 Myr (depending on analysis), *Kikihia* lineages have accumulated in a notably steady manner. Lineage-through-time plots (Fig. 4) and diversification rate contrasts suggest little effect of the Pleistocene boundary and only a slight effect of the mid-Pleistocene onset of 100 ky climate cycles in NZ around 1.2 Ma (Carter, 2005) (see calculation B, Fig. 4), an overall conclusion strengthened by the failure of the data to reject a constant-rate pure-birth model as judged by the γ statistic of Pybus and Harvey (2000).

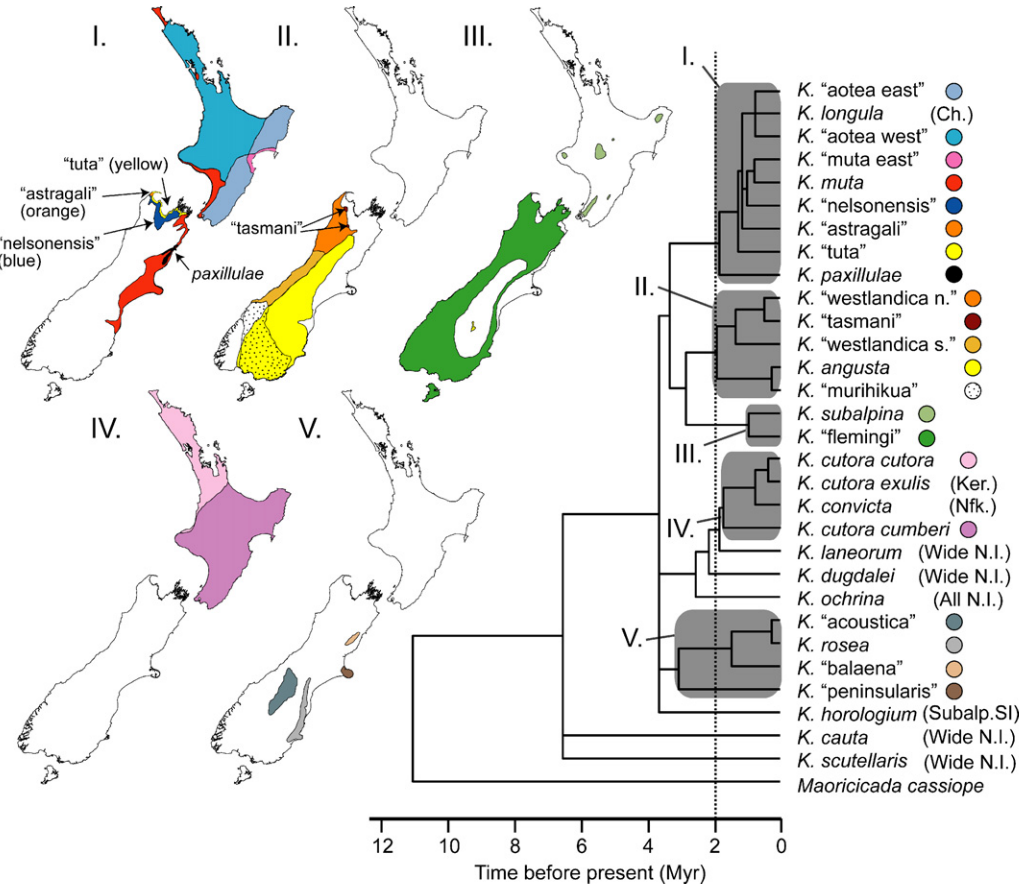

Fig. 5. Restriction of geographic sympatry to *Kikihia* lineages with divergence times greater than approximately 2 Ma. Taxa grouped within a given shaded box are mutually allopatric or parapatric and form hybrid zones upon contact. Taxa from different shaded boxes or unboxed lineages are commonly sympatric. Vertical dotted line approximates the "sympatry threshold". Simplified geographic distributions of species within each boxed group I–V are shown in accompanying maps. Distributions of the remaining species are summarized verbally as follows: *K. horologium* is found in subalpine areas of South Island. *K. longula* is found only on the Chatham Islands. *K. cutora exulis* is found only on the Kermadec Islands. *K. convicta* is found only on Norfolk Island. Species labeled "Wide N.I." are widely distributed on North Island and frequently found in sympatry. The chronogram was obtained from a relaxed-clock dating analysis (Analysis E, see Methods) using the *Kikihia* mtDNA dataset.

Because visual inspection of the LTT plot (Fig. 4) suggests a possible increase in diversification rate around 3.7 Ma followed by a later minor decrease around 1 Ma, additional tests were conducted to determine if the overall nonsignificant γ statistic was caused by counterbalancing trends in different time periods (Nee, 2001). The γ statistic over the period 4–1 Ma was positive instead of negative, but still nonsignificant ($\gamma = 1.22$). A significant deviation from the constant-rates model ($\gamma = 1.99$, a marginally significant increase in rate over time) was found when the period 8.8–1.0 Ma was considered, so the apparent shift around 4 Ma may be real. Nonetheless, *Kikihia* diversification has occurred with little statistically detectable change in rate, especially since 3–4 Ma.

4.3. Extinction patterns and lineage-through-time plots

There is some support in the dating results for the hypothesis that early Pliocene (5 Ma) acceleration in growth of the Southern Alps initiated diversification in *Kikihia*. The analysis with the preferred priors and calibrations (Analysis E) dated the first split to 6.56 Ma, and two of the six analyses estimated initial divergences younger than 5 Ma. However, no further splits appear until <4 Ma when the LTT plot curves upward, suggesting that either mountain-building had a delayed effect on *Kikihia* or that other processes obscured its effects. Extinction is a real possibility, given the considerable changes in NZ habitats that have occurred over this time frame, but it is one that is difficult to test with LTT plots. A constant background rate of extinction, combined with a constant speciation rate, yields a plot that is at first steady, then upwardly curving to the present (e.g., Fig. 3 in Nee et al., 1994)—not downward-sloping or linear to the present as in Fig. 4—so a constant background extinction rate is not supported.

Alternative explanations for the apparent mid-Pliocene increase in diversification rate are (1) a simple increase in speciation rate or (2) a biased episode of extinction (as opposed to a constant background extinction rate). The latter could have occurred as a single event before 4 Ma, or later if ongoing extinction disproportionately influenced earlier-diverging lineages (similar to the "clade extinction" of Harvey et al., 1994). The lineages that diverged before 4 Ma are today represented by *K. cauta* and *K. scutellaris*, two of only three species in the genus that inhabit dense lowland forests that likely predominated in early Pliocene New Zealand. Most descendants of the main radiation inhabit "newer" habitats (grass, scrub, forest edges) that appeared or expanded during the Pliocene and Pleistocene. Thus, the shallower slope prior to 4 Ma could be explained if forest-dwelling species were lost preferentially through habitat attrition during the late Pliocene and Pleistocene when forests retreated to the northern regions of the North and South Island (NZ-INTIMATE Project; Barrell et al., 2005).

The above interpretations are necessarily tentative because a given LTT plot can be explained by many different combinations of speciation and extinction rates (Kubo and Iwasa, 1995), particularly if those rates change over time (see also Nee, 2006). Many lineages, once formed, apparently did not undergo additional speciation events—suggesting that birth and/or death rates may have varied over time. Furthermore, conflicts between the EF-1α and mtDNA trees mean that the timing of some splits remains in doubt. Fortunately, a small number of phylogenetic rearrangements should not have a large effect on our understanding of the overall diversification pattern.

4.4. Other Plio-Pleistocene radiations in New Zealand

Many studies of evolutionary radiations in New Zealand have sought correlations between key historical events and the initial diversification of major groups, usually genera. Several have identified the climatic and (especially) orogenic processes of the late Miocene to late Pliocene as most important for diversification (e.g., freshwater crayfish: Apte et al., 2007; alpine *Maoricicada*: Buckley and Simon, 2007; wrens: Cooper and Cooper, 1995; alpine buttercups: Lockhart et al., 2001; weta: Trewick and Morgan-Richards, 2005; Carmichaelia: Wagstaff et al., 1999; Stylidiaceae: Wagstaff and Wege, 2002), while comparatively few have found evidence for Pleistocene-age radiations (*Cyanoramphus* parakeets: Chambers et al., 2001; whipcord *Hebe*: Wagstaff and Wardle, 1999). In a wide-ranging genetic survey of ten endemic invertebrate genera, Trewick and Wallis (2001) found almost no Pleistocene-age divergences. However, their sampling scheme was designed to determine the general timing of diversification of each taxonomic group rather than describe each case in detail. It is possible that complete taxon sampling would substantially increase the amount of Pleistocene-age divergence in some of the groups analyzed.

Although most extant *Kikihia* lineages date to the Pleistocene, we find no evidence that Pleistocene processes (e.g., stronger cycles of population expansion and contraction; Winkworth et al., 2005) have been more important that those of the late Pliocene. If such processes have specifically contributed to NZ cicada speciation as previously proposed (Fleming, 1975, 1979), the expected increase in *Kikihia* diversification has been offset by matching decreases in other species-generating mechanisms, or by increased extinction. However, some studies (e.g., Hughes and Eastwood, 2006; Turgeon et al., 2005) have shown apparently strong associations between Pleistocene climates and diversification rates, and a general theory is still needed to explain these disparate patterns.

4.5. Comparison with the NZ alpine genus Maoricicada

The divergence time estimates obtained for *Kikihia* are similar to those obtained for the well-sampled alpine cicada genus *Maoricicada* (Buckley and Simon, 2007; Hill, 2005) which, like *Kikihia*, appears to have radiated recently. However, only about 20% of *Maoricicada* species originated in the Pleistocene, and the main alpine *Maoricicada* radiation was dated (using mtDNA and nuclear EF-1α) to 4.5–4.8 Ma (respectively) as opposed to 3.65 Ma for the main radiation in *Kikihia* (Analysis E), suggesting a more important role for early Pliocene processes in *Maoricicada*. Encouragingly, our best analysis yielded remarkably similar estimates to those found by Buckley and Simon for the *Kikihia-Maoricicada* divergence (ca. 11 Ma) and the initial divergence of the focal genus (6.5–6.9 Ma for *Maoricicada*, 6.6 Ma for *Kikihia*). Many of the *Maoricicada* species inhabit high elevation, tundra-like habitats, and the earliest splits within the genus apparently occurred between low- and high-elevation stem lineages on the South Island (Buckley and Simon, 2007). In contrast, no *Kikihia* inhabit high-elevation habitats, only three lineages are subalpine, and the oldest splits seem to have involved lower-elevation forest-dwelling lineages now found only on North Island. The characteristics and distributions of the early *Maoricicada* lineages may have made them more adaptable to the South Island mountain-building and habitat change that accelerated in the early and mid-Pliocene, while the *Kikihia* ancestors (with possibly a more northern historical range) did not radiate until the accelerating deterioration of climate in the mid-Pliocene.

4.6. Geography and the later stages of speciation in Kikihia

During speciation, a new population-level lineage gradually acquires evolutionary independence (de Queiroz, 1998) from related lineages through a combination of ecological, physiological, and reproductive divergence. The ability to coexist in sympatry may indicate that ecological and reproductive divergence has brought two lineages to a final stage of evolutionary independence (although genetic isolation may remain incomplete indefinitely; e.g., Grant et al., 2005). Many of the *Kikihia* taxa discussed here have parapatric distributions with contact, hybridization (as indicated by song and DNA evidence; unpublished data), and potential introgression of genes. For allopatric taxa, there is little or no chance for natural hybridization and this provides us with no clues as to degree of differentiation, however with naturally parapatric and sympatric taxa we see a repeated trend. When the spatial relationships are mapped onto the *Kikihia* chronogram (Fig. 5), a "sympatry threshold" appears around 2 Ma. Nearly every species is parapatric or allopatric to every other taxon from which it has diverged during approximately the past 2 million years, while many lineages that diverged earlier are broadly sympatric, with no evidence of hybridization (Fleming, 1975; Lane, 1995, pers. obs.). While correlations between divergence time and degree of sympatry have been demonstrated before (e.g., Barraclough and Vogler, 2000; Jordal and Hewitt, 2004; Losos and Glor, 2003), the concordance of the sympatry threshold across different *Kikihia* lineages makes the present study especially interesting.

As might be expected, the "time to speciation" as judged by the appearance of sympatry (a late stage) is at the upper bound of that estimated by Avise et al. (1998) (1–2 Myr) and McCune and Lovejoy (1998) (0.8–2.3 Myr) based on lineage splitting rates. Furthermore, the complete absence of sympatric species with divergence times under 2 Ma could indicate that this aspect of differentiation has slowed during the Pleistocene, even while diversification has remained generally constant. Climate proxies (see Fig. 4) suggest that directional climate change (cooling) may have slowed in the mid-Pleistocene, with subsequent changes affecting mainly the amplitude and periodicity of climate cycling. Perhaps sustained climate-driven changes are especially important for the ecological divergence necessary for new species to become able to coexist.

Variation in species concepts applied in different areas of biology, as well as differences in the ease of detecting cryptic diversity across groups, should be considered in deriving general conclusions about the timing and causes of organismal diversification. With all recent diversification in *Kikihia* leading to mutually parapatric or allopatric taxa, and with most of those actually or potentially hybridizing in contact zones, strict application of the biological species concept would cause many of these taxa to be lumped together and thereby remove most evidence of Pleistocene-age diversification. Similarly contrasting conclusions have been reached by teams of investigators applying different taxonomic criteria to the analysis of avian diversification (e.g., Avise and Walker, 1998; Johnson and Cicero, 2004; Klicka and Zink, 1997). Furthermore, it is possible that the high levels of Pleistocene-age "intraspecific" population-genetic divergence documented for groups such as NZ cockroaches (Chinn and Gemmell, 2004) and weta (Trewick and Morgan-Richards, 2005) may sometimes indicate diversification processes analogous to those driving the evolution of cryptic song species in cicadas.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.ympev.2008.05.007.

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